Learning Objectives

At the end of this chapter, students are expected to know

- the limitations of right triangle trigonometric ratios as they involve acute angles.
- the necessity for the study of radian measure of an angle and its advantage over degree measure.
- how unit circle is used to define trigonometric functions of real numbers.
- various trigonometric identities, their relationships and applications.
- the principal solution and general solution of a trigonometric equation.
- how to solve trigonometric equations.
- law of sines, law of cosines in triangles and their applications in real life situations.
- how to solve an oblique triangle using law of sines and law of cosines.
- application of Heron's formula and how to compute area of a triangle without finding its altitude.
- the existence of inverse trigonometric functions and their domains and ranges.

Let us recall the basics of trigonometric ratios using acute angles and their properties, which were discussed in earlier classes.

3.2 A recall of basic results

In earlier classes, we have learnt trigonometric ratios using a right triangle and proved trigonometric identities for an acute angle. One wonders, how the distance between planets, heights of Mountains, distance between far off objects like Earth and Sun, heights of tall buildings, the speed of supersonic jets are measured or calculated. Interestingly, such distances or heights are calculated applying the trigonometric ratios which were derived for acute angles. Our aim is to develop trigonometric functions defined for any real number and use them in all branches of mathematics, in particular, in calculus. First, let us recall the definition of angle and degree measure of an angle.

3.2.1 Angles

The angle AOB is a measure formed by two rays OA and OB sharing the common point O as shown in the Figure 3.1. The common point O is called the **vertex** of the angle. If we rotate the ray OA about its vertex O and takes the position OB, then OA and OB respectively are called the initial side and the terminal side of the angle produced. An anticlockwise rotation generates a positive angle (angle with positive sign), while a clockwise rotation generates a negative angle (angle with negative sign).

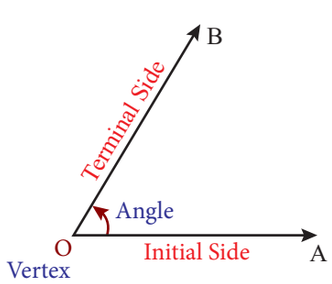


Figure 3.1

One full anticlockwise (or clockwise) rotation of OA back to itself is called one **complete rotation or revolution**.

3.2.2 Different Systems of measurement of angle

There are three different systems for measuring angles.

(i) **Sexagesimal system**

The **Sexagesimal system** is the most prevalent system of measurement where a right angle is divided into 90 equal parts called Degrees. Each degree is divided into 60 equal parts called **Minutes**, and each minute into 60 equal parts called **Seconds.**